



Information Systems Research

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To cite this article:

Yash Babar, Gordon Burtch (2020) Examining the Heterogeneous Impact of Ride-Hailing Services on Public Transit Use. Information Systems Research

Published online in Articles in Advance 02 Apr 2020

. <https://doi.org/10.1287/isre.2019.0917>

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Examining the Heterogeneous Impact of Ride-Hailing Services on Public Transit Use

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Received: March 16, 2018

Revised: February 11, 2019; September 19, 2019

Accepted: November 14, 2019

Published Online in Articles in Advance:
April 2, 2020

<https://doi.org/10.1287/isre.2019.0917>

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Abstract. We examine the impact that ride-hailing services have had on the demand for different modes of public transit in the United States, with a particular focus on understanding heterogeneity in the effects. We assess these effects using a panel data set that combines information on public transit utilization (from the Federal Transit Administration) with information on ride-hailing providers' staggered arrival into different locations, based on public press releases and newspaper reports. Our analysis indicates that, on average, ride-hailing services have led to significant reductions in the utilization of city bus services while increasing utilization of commuter rail services. These average effects are also subject to a great deal of contextual heterogeneity, depending on the size of the local population, rates of violent crime, weather, gas prices, transit riders' average trip distance, and the overall quality of public transit options. We demonstrate the robustness of our findings to alternative model specifications. Our findings contribute to the prior literature on technology substitution and complementarity and suggest explanations for contradictory findings that have been reported on ride-hailing's influence on public transit demand. We also offer useful insights for policymakers, highlighting the nuanced implications of ride-hailing services for different transit operators, depending on the local context.

History: Anandasivam Gopal, Senior Editor.

Supplemental Material: The online appendices are available at <https://doi.org/10.1287/isre.2019.0917>.

Keywords: Uber • Lyft • public transit • difference-in-differences • ride-hailing • transportation

1. Introduction

With more than 10 billion rides under its belt, the ride-hailing service Uber now stands as a relatively mature private operator in the urban mobility market, warranting systematic studies of how it changes the way we move. Several transit agencies have begun to respond to the disruption wrought by such services, engaging in direct partnerships with ride-hailing providers (Brustein 2016, Grabar 2016) or subsidizing citizens' use of ride-hailing to cover the "last mile" of transit.¹ To understand the potential value of these responses, it is important to first develop a systematic understanding of the impact that ride-hailing services have had on public transit use. A few initial works have examined this question (Clewlow and Mishra 2017, Hall et al. 2018, Nelson and Sadowsky 2018). Recognizing that every city and transit operator operates in its unique environment, we revisit the question, to understand various sources of heterogeneity that underlie the average effects. Formally, we seek to address the following research questions: How does access to ride-hailing services affect the demand for specific modes of public transit? What factors amplify or attenuate these effects?

We adopt a technology substitution/complementarity lens to motivate our analysis. The value of these on-demand transportation services, whether as substitutes or complements for public transit offerings, can vary a great deal across locations. In some cities, public transit may fall short in many ways, leaving gaps for ride-hailing to fill: riders may feel unsafe walking to or waiting at a bus shelter, or they may want to avoid inclement weather; riders in different locations may contend with higher or lower gas prices, or they may benefit (suffer) from superior (worse) public transit options. For ride-hailing providers, understanding these gaps in incumbent public transit offerings is important for determining potential demand for ride-hailing in any given location. An understanding of these factors may also inform stakeholders, for example, transit agency operators or municipal governments, and shape their decisions about how to move forward. Finally, some of the aforementioned factors may be influenced by government investment—for example, the government might increase transit service frequency or invest in crime prevention—suggesting important levers at policymakers' disposal for shaping their cities' experience with ride-hailing.

In the following sections, we begin by discussing the prior literature that has considered the effect of ride-hailing on public transit demand. Drawing on the technology substitution and complementarity literature, we highlight the potential for heterogeneity in this new transit technology's effects and point to this as a gap in the existing ride-hailing literature. We then document a set of plausible moderating factors that have seen independent consideration across studies of ride-hailing services in several different fields of study. Finally, we empirically evaluate the direct and moderating relationships in question, with a particular focus on city bus and commuter rail services.² We employ a panel data set that combines the schedule of ride-hailing services' entry across locations in the United States between January of 2012 and July of 2018 with data on public transit ridership statistics reported to the Federal Transit Administration (FTA) by individual transit agencies.

We estimate that ride-hailing platforms reduce utilization of city bus services by approximately 1.29%, on average, yet expand the demand for commuter rail services by approximately 2.96%, on average. We observe no statistically significant average effects on the utilization of other transit modes (although we do observe heterogeneous effects that are presented in Online Appendix E). Subsequently focusing on city bus and commuter rail services, we expand our specification to incorporate the various potential sources of heterogeneity—namely those associated with rates of crime, frequency of precipitation, fluctuations in gas prices, the size of a city's labor force, and the distance traveled by an average user of the public transit mode. Upon doing so, we document a wide array of heterogeneous effects. We also assess the robustness of our results in a variety of ways, considering alternative model specifications and identification strategies.

2. Related Work

It is often assumed that new technologies will tend to substitute existing ones, despite many examples to the contrary (Young 1993). For instance, although new technologies may substitute for existing technologies or human labor when it comes to firms and individual workers, under some conditions (e.g., with the right organizational design and the appropriate level of multitasking, respectively), the two have been shown to act as complements (Hitt and Snir 1999, Aral et al. 2012). Thus, to arrive at an appropriately nuanced understanding of the relationship between new digitally enabled peer-to-peer offerings and incumbent services, it is important to consider various factors that characterize the context of the interaction because these will determine whether and to what degree the interaction is one of substitution or complementarity.

In the literature examining such relationships for other two-sided sharing/gig economy platforms, Zervas et al. (2017) is noteworthy for demonstrating that the entry of Airbnb significantly reduces the demand for traditional hotels. Similarly, Zhang et al. (2018) provides evidence that ride-hailing and home-sharing services are complementary to one another; they demonstrated that the exit of ride-hailing services from a local area results in a significant decline in Airbnb demand. In both cases, patterns of substitution or complementary were noted to depend on other factors. Zhang et al. (2018) observe that the complementary relationship is particularly strong for Airbnb listings located further from public transit, whereas Zervas et al. (2017) observe that demand substitution is particularly manifest for hotels that do not cater to business travelers. Our work expands on this line of inquiry, considering the interaction between the availability of ride-hailing services and the demand for public transit modes. In doing so, we consider a variety of contingent factors (detailed below) that have the potential to dictate the pattern of interaction between ride-hailing demand and the demand for public transit services.

Exploring sources of heterogeneity, our work builds on some recent studies in economics, transportation, and management that have examined on-demand ride-hailing services. Studies have documented relationships between ride-hailing service availability/use and entrepreneurial activity (Burtch et al. 2017), rates of violent crime (Dills and Mulholland 2018, Park et al. 2016), inclement weather (Brodeur and Nield 2018), and urbanization and land-use mix (Alemi et al. 2018). Within the area of transportation, specifically, past work has examined the effect of ride-hailing services on the incidence of drunk driving (Brazil and Kirk 2016, Greenwood and Wattal 2017), the volume of private vehicle purchases (Gong et al. 2017, Ward et al. 2018), and traffic congestion (Li et al. 2016, Erhardt et al. 2019).

Of greatest relevance to the present study is the small set of work that has examined the relationship between ride-hailing and public transit. By surveying urban transit riders, for example, past work has found that individuals often opt to employ ride-hailing services even in the presence of cheaper transit options if they anticipate a large reduction in trip duration (Rayle et al. 2014, Murphy 2016), a perception that may depend on the features of a particular public transit mode (Clewlow and Mishra 2017). Other recent work has leveraged archival data on aggregate public transit use to study these relationships. Hall et al. (2018) conclude that ride-hailing services produce a net increase in public transit demand, particularly in populous cities. In contrast, Nelson and Sadowsky (2018) conclude that in the long term,

public transit tends to be displaced, particularly when two ride-hailing operators are operating in tandem.

Our study contributes to this literature by offering a nuanced consideration of moderating factors that may combine to determine patterns of substitution or complementarity between ride-hailing and public transit. In the next section, we describe the set of factors we consider in our analyses, with supporting rationale; we consider a combination of transit features, preestablished transit rider behaviors, and the transit operating environment.

3. Sources of Heterogeneity

Each city, and thus its public transit system, is relatively unique. Cities and transit modes vary widely in terms of their physical environment and operating characteristics. Given this, we explore a variety of factors that speak to the potential for substitution and complementarity between ride-hailing and public transit service modes, grouping these factors into (1) transit service features, (2) transit rider behavioral/perceptual factors, and (3) environmental factors.

3.1. Transit Service Features

Certain transit modes are predisposed toward a complementary (rather than substitutive) relationship with ride-hailing services in at least two respects. First, prior work notes that public transit modes that benefit from a separate *right of way* (ROW) have distinct advantages over services that must share the road with personal vehicles (Dittmar and Ohland 2012). Both commuter rail and subway services are prime examples of this. Commuter rail services operate over individual tracks or rails that are inaccessible to private vehicles. These services provide commuters with access to densely populated areas while avoiding roads that may be congested with private vehicle traffic. Commuter rail services thus offer a unique degree of convenience, affordability, and efficiency over road-based transit options (Vuchic 2005). In contrast, other public transit modes tend to operate exclusively via road, in direct competition with private vehicle traffic, most notably city bus services.³

Second, prior literature suggests that ride-hailing services have the potential to expand the use of public transit modes by supplying last-mile connectivity (Clewlow and Mishra 2017). Commuter public transit options typically service suburban residents distributed over a relatively wide geographic radius (Vuchic 2005). As a result, the last mile of transit is typically not delivered directly by the public transit operator; rather, riders must find their way to and from the transit stop. This last mile of transit may involve the use of a private vehicle—for example, driving and parking a car over the course of the workday (park-and-ride; Noel 1988) or being dropped off by a third

party (kiss-and-ride; Schank 2002). Ride-hailing services have the potential to complement commuter rail services by helping transit riders cover the last mile with relative comfort and ease. City bus services, in contrast, are typically operated with an eye toward the greatest coverage of residential areas, stopping at a relatively large number of locations, every few blocks. As such, the potential for last-mile connectivity benefits is significantly reduced with city bus services.

We account for this form of heterogeneity by estimating ride-hailing's effects on the utilization of each public transit mode. Specifically, we contrast the effects on city bus services (that do not maintain a unique right-of-way and that stand to benefit less from last-mile connectivity) with commuter rail services (that do maintain a unique ROW and that stand to benefit more from last-mile connectivity).⁴

3.2. Behavioral/Perceptual Factors

The next set of factors we consider pertains to consumer/rider behavior and perceptions. These include factors that indicate how a particular mode of transit is used or perceived in a location.

3.2.1. Average Trip Length. Across different transit systems, depending on the network connectivity, frequency, and local geography, there can be significant variation in the distance travelers seek to cover via different public transit modes. For example, the average ride taken on the city bus in Ames, Iowa, is just 1.6 miles in distance. In contrast, the average trip spans more than 12 miles in Pittsburgh, Pennsylvania. Thus, in terms of the way riders employ the service, Ames tends to accommodate shorter trips, whereas Pittsburgh tends to accommodate lengthier trips. This suggests that ride-hailing would be a more viable replacement in the case of Ames than in the case of Pittsburgh, given the greater cost differential in the latter case between ride-hailing and bus services. In geographies where city bus services tend to be used for long-haul transit, we would expect the absence of a relationship, or perhaps a shift toward complementarity—for example, riders may employ ride-hailing services to cover the trip to a bus station.

3.2.2. Transit Quality Perceptions. One obvious factor that may moderate the effect of ride-hailing service entry on public transit demand is the preexisting quality of the public transit options. Providers that service larger coverage areas, operate with greater frequency, and experience fewer outages should be less likely to be displaced by ride-hailing services. Additionally, any possible complementarity between ride-hailing services and public transit options—for example, coverage of the last mile, an effective

expansion of the transit systems coverage area—should be increasing in the quality of the public transit system. Conversely, lower-quality transit options should be more likely to be displaced by ride-hailing options and less likely to benefit from complementarities.

3.3. Operating Environment

We also consider the local operating environment, identifying and discussing several factors that have the potential to determine ride-hailing effects on public transit demand, including population, unemployment, gas prices, crime, and weather.

3.3.1. Population. Large urban areas in the United States are characterized by high population density, denser central commercial districts, high-frequency transit options, and more expansive coverage of the residential population (Taylor et al. 2009). In such cities, a greater proportion of residents' travel needs are commonly addressed by public transit options. By contrast, in smaller towns and cities, driving remains the primary travel mode because public transit tends to be less frequent in its operation and fewer modes of transit tend to be offered. At the same time, larger cities host a larger population of potential drivers and consumers and thus are more likely to exhibit a critical mass of participants in the ride-hailing market, implying stronger network effects and ultimately a greater value proposition of ride-hailing relative to public transit. Notably, the recent work by Hall et al. (2018) has documented this particular source of heterogeneity in the effect of ride-hailing services on the demand for public transit. Those authors observed that the effects of ride-hailing on transit demand manifest more strongly in more populous cities. Consistent with Hall et al. (2018), we expect to observe that the effects of ride-hailing will be stronger in more populous cities.

3.3.2. Gas Prices. The price of gasoline is an important consideration in transit mode choice (Wang and Skinner 1984, Lane 2010, Iseki et al. 2014) not only for public transit options but also for ride-hailing services as well.⁵ A great deal of anecdotal evidence indicates that although changes in gas prices are not directly factored into the fares charged by ride-hailing services, they nonetheless have an indirect effect on the value of ride-hailing services for consumers in two ways. First, as gas prices increase, because ride-hailing intermediaries do not adjust driver wages in response, driver profit margins are reduced. This reduces drivers' interest in supplying services, which leads to fewer cars on the road and thus increases in consumer wait times.⁶ Second, because the dynamic pricing algorithms operated by ride-hailing intermediaries *do* respond to imbalances in supply

and demand, as the number of drivers operating reduces, the frequency of price surges increases.

Given this situation, we might expect that the value proposition of ride-hailing thus reduces with increases in gas prices, whereas that of public transit options remains relatively fixed in the short term. Specifically, for public transit modes that face direct competition from ride-hailing services, such as city buses, we would expect them to experience weaker displacement as ride-hailing services become more expensive alternatives. Conversely, for transit options that are potential complements to ride-hailing services, such as commuter rail, we would expect the growth in utilization associated with ride-hailing entry to be attenuated as ride-hailing services become more expensive.

A final note is warranted about the effect of gas price fluctuations on the delivery of public transit options. Like ride-hailing intermediaries, public transit agencies rarely revise their fare schedules and thus tend not to react to short-run increases or decreases in gas prices. However, unlike ride-hailing providers, public transit operators maintain direct control over transit supply; thus they also continue to operate transit vehicles at regular frequencies regardless of fluctuations in operating costs.

3.3.3. Crime and Safety. Crime on transit vehicles and at transit stops strongly influences riders' perceptions of safety and thus the decision to use transit (Pearlstein and Wachs 1982). Riders have been known to avoid certain transit access points and times of travel out of a fear of being victimized. Whereas crimes such as vandalism and drunkenness affect the quality of the transit experience, crimes that threaten the safety of transit riders may reduce transit demand generally (Pearlstein and Wachs 1982, Loukaitou-Sideris et al. 2002), creating an opportunity for ride-hailing to relieve friction. Supporting this idea, the consumption of ride-hailing services has recently been shown to correlate with local rates of physical assault (Park et al. 2016, Dills and Mulholland 2018). One plausible rationale for how violent crime may moderate the effect of ride-hailing services on the demand for city bus and commuter rail services is thus as follows. If street-level crime is high, then a rider may be more likely to entirely replace cheaper, short-haul transit trips with ride-hailing options to avoid a short walk at the beginning/end of the trip or waiting on the street at a bus stop. Conversely, under higher crime, a rider may be more likely to take a long-haul transit trip (which will typically require last-mile connectivity) if he or she knows that there is a safe way to complete the last mile that does not involve waiting alone for a bus or walking to the parking lot. With this rationale, violent crime is expected

to amplify the displacement of city bus services and amplify ride-hailing's expansion of commuter rail services.

3.3.4. Weather. The transit literature has repeatedly demonstrated that weather is an important predictor of transit mode choice (Arana et al. 2014). Inclement weather can have a variety of effects on this choice. In the presence of inclement weather, travel in a private vehicle, via road (whether a personal vehicle or a ride-hailing service), may be perceived as strictly more convenient because it allows the rider to avoid exposure to the elements. This applies not only to precipitation (Brodeur and Nield 2018) but also to extreme temperatures (Arana et al. 2014). To this point, past work has observed that rail ridership increases with riders' attempts to avoid over-the-road congestion and risky driving (Guo et al. 2007). However, in general, the effects of inclement weather on transit mode choice are quite complicated for several reasons.

First, the weather can shift quickly, from one hour to the next. As a result, the dynamic pricing mechanisms employed by ride-hailing services are responsive to sudden spikes or shifts in demand that manifest in response to the weather. For example, in rainfall or snowfall, price surges are common as the demand for ride-hailing services temporarily rises. These price increases are intended to balance supply and demand,

in part by increasing the volume of drivers but also by reducing consumption. As such, the dynamic pricing mechanisms may countervail any heterogeneity altogether.

Second, the effects may depend on the exact form of inclement weather. Whereas rain has the potential to disrupt travel instantaneously, snow and ice can accumulate and persist over longer periods, affecting transit in a persistent fashion, hours or days after actual snowfall has taken place. Further, whereas rain and snow can lead to on-road congestion and riskier driving conditions, this generally is not the case with extreme temperatures. This latter point suggests that extreme temperatures should lead to an increased preference for private vehicles—for example, ride-hailing services. The effects of precipitation, however, are not altogether clear. As such, we turn to the data to understand these relationships.

4. Methods

4.1. Data

We next describe our measures and data (see Table 1 for a summary). Individual transit agencies report monthly estimates of agency ridership and vehicle operation statistics to the FTA. Each agency operates within a Department of Transportation–designated Urbanized Area (UZA), which serves as our geographic unit of analysis. The primary metric of demand that agencies and federal authorities focus on

Table 1. Summary of Measures and Data Sources

Variable	Description	Data source
$Utilization_{i,t}^m$	Monthly total utilization of mode m in geography i in month t . Calculated as the total number of unlinked passenger trips (UPT) for the mode m in the given month and city divided by the total number of vehicle revenue hours (VRH).	National Transit Database (Federal Transportation Administration 2018a)
$AvgTripLength_{i,t}^m$	Length of the average trip (miles) in geography i for a given mode m as reported by the transit agency.	
$RideHailing_{i,t}$	Binary indicator of presence of any Uber service in geography i in month t .	Uber press blog, news websites
$LaborForce_{i,t}$	Total size of the workforce in geography i in month t .	Bureau of Labor Statistics (U.S. Census Bureau 2018)
$Unemployment_{i,t}$	% rate of unemployment in geography i in month t .	
$GasPrice_{i,t}$	Average price of gas per liter in U.S. dollars in geography i in month t .	Gas prices by PADD (U.S. Energy Information Administration 2018)
$NumHotDays_{i,t}$	Number of days where the temperature is high ($\geq 30^\circ\text{C}$) in geography i in month t .	NOAA Global Surface Summary of the Day (National Climatic Data Center 2018)
$NumColdDays_{i,t}$	Number of days where the temperature is low ($\leq 0^\circ\text{C}$) in geography i in month t .	
$NumRainyDays_{i,t}$	Number of days where the daily rainfall exceeded 1 inch in geography i in month t .	
$NumSnowDays_{i,t}$	Number of days where the daily on-ground snow accumulation exceeded 1 inch in geography i in month t .	
$ViolentCrime_i$	Total count of reported crimes involving murder, rape, robbery, and aggravated assault in 2014 in geography i per thousand people.	FBI's Uniform Crime Report (U.S. Department of Justice 2014)
$TransitQuality_i$	AllTransit's patented quality score (out of 10) capturing the quality of all public transit offerings in a given geography i .	AllTransit (Center for Neighborhood Technology 2018)

Note. Variables defined for mode m in city i in monthly period t .

is utilization, calculated as the volume of unlinked passenger trips (UPTs) divided by cumulative hours of service across all vehicles operated in a given month: vehicle revenue hours (VRHs). If multiple transit agencies were operating the same transit mode in a single UZA, we aggregated the data across agencies to arrive at our final cumulative measure of UZA mode-specific *utilization* over time. The underlying data are published for public consumption at the National Transit Database (NTD) website. Other measures published by the NTD about each transit mode, operated by each agency, include average trip length (ATL), average fare per trip (AFPT), average operator trip cost, and agency service area, among other agency-level statistics. This information is available for 2,276 transportation agencies over our period of study. The 2017 NTD agency report contains values reported by transit agencies servicing 379 urbanized areas. Our observation window begins in January of 2012 because the NTD changed its reporting standards at that time, reclassifying certain bus and rail services around the United States to distinguish, most notably, between commuter and local services. Thus the monthly totals per each agency before and after 2012 are not comparable because the measures pertain to fundamentally different populations of transit agencies and definitions of modes.

As noted earlier, our focus is on city bus and commuter rail services.⁷ According to the NTD data reporter's manual (Federal Transportation Administration 2018b), city bus services are defined by the FTA as fixed-route services powered by motor and fuel contained within a vehicle. These services are characterized by relatively short distances between stops, for service primarily operating within denser areas. Commuter rail services are defined by the FTA as rail services operating on either old freight railways or tracks that are shared with freight railways, Amtrak, or both. The service is characterized by relatively long distances between stops, for service primarily connecting a central city with outlying suburbs and cities. The service may be either diesel or electric powered and usually has grade crossings with roadways.

4.1.1. Key Outcome and Independent Variable. Our outcome of interest is public transit *utilization*—that is, the volume of ridership normalized with respect to cumulative vehicle revenue hours. This measure is reported as the primary indicator of transit productivity to the FTA by transit agencies.⁸ Our key independent variable of interest is a binary indicator of ride-hailing service availability. As noted earlier, we initially focus on the entry of Uber rather than ride-hailing providers in general because Uber is by far

the largest provider of ride-hailing services in the United States and was the first entrant in more than 90% of locations. In cities where Lyft was the first entrant, Uber followed within just a few months. Accordingly, for simplicity's sake, we consider the Uber-availability indicator to be representative of access to ride-hailing services in a geographic area generally. Of course, this has implications for how we interpret the effect estimates from our regressions.

We obtained the schedule of Uber's entry dates across cities directly from Uber. In a handful of cases, discrepancies were observed between the Uber-provided schedule and entry dates collected from local news reports. In such cases, we employed the earlier of the two dates. Our key independent variable identifies the first introduction of any Uber service variant in the location (whether Uber X, Uber Black, etc.). In ~80% of the locations in our sample, Uber X was the first service offered. A detailed list of Uber entry events is provided in Online Appendix I.

4.1.2. Population and Unemployment Rate. We proxy a city's population with a measure of the local labor force size (the sum of employed and unemployed persons) in each UZA, by month, from the Bureau of Labor Statistics (BLS). The size of the labor force is a reliable proxy for the size of the broader urban population while also being recorded frequently, enabling the capturing of seasonal variation in the local commuter population. In contrast, general population data are recorded relatively infrequently by the U.S. Census Bureau, typically on an annual basis. We also collect data on the local unemployment rate by month, again from the BLS. Employment is an important control variable because it is a prime determinant of transit ridership (Taylor and Fink 2003). This is because a significant portion of transit demand derives from work commuters (Hendrickson 1986). Further, employment is a key factor underlying disposable income.

4.1.3. Gas Price. The U.S. Energy Information Administration (EIA) provides average gasoline prices for each Petroleum Administration of Defense District (PADD), defined as regions of the United States that comprise multiple states. The measure is derived from the weekly gas prices at approximately 130,000 gas stations identified by the EIA as of 2016.⁹ The final measure, recorded in U.S. dollars per gallon, is calculated as a weighted average across all gas stations in a given PADD, with weights based on the volume of sales at each gas station to account for the most prevalent prices experienced by consumers. We map these PADD-based monthly gasoline price measures to UZAs.

4.1.4. Transit Quality. To capture the overall quality of a transit system in a given city, we use the Transit Score obtained from AllTransit.com. This provides a measure of public transit quality aggregated over all modes of service in an urbanized area. This score, which ranges from 0 to 10, reflects (1) the degree to which public transit in a city enables a work commute, (2) the frequency of transit operation in a city, (3) the percentage of population serviced by public transit in a city, and (4) the proportion of locations in a city that are within walkable distance of transit stops, along with a number of other related parameters obtained from data reported by each agency through the Generalized Transit Feed Specification.¹⁰ This measure is a reasonable reflection of the quality of public transit services in a city, which accounts for the demography, geography, and timing of service offerings (Litman 2008). This measure also has a very high correlation (0.82) with another patented measure of transit quality that has been used in prior studies in the transportation literature (Duncan et al. 2013)—namely the TransitScore (Lerner et al. 2017), reported by WalkScore.com. However, unlike the latter measure, the AllTransit score we use is available for a much larger set of cities. This score was reflective of quality as of August 2016.

4.1.5. Crime and Safety. We collected the total number of violent crimes per capita in each city as of 2012 from the FBI's Uniform Crime Reporting system. Crimes listed in this measure include murder, rape, armed robbery, and assault. We sum these crime volumes across all counties in an urbanized area.

4.1.6. Average Trip Length. Average trip length is reported directly by transit agencies, per transit mode, per location, in the NTD. We use this variable to measure the mean distance traveled by the

average consumer (per trip) of the transit mode in a given location.

4.1.7. Weather. We collected weather data from the National Oceanic and Atmospheric Administration (NOAA). We calculate monthly measures of the number of hot days, cold days, rainy days, and snowy days per month based on daily historical data. For any given UZA, we leverage weather reports recorded at the nearest active reporting weather station, defined as that which is geographically closest to the center of the largest city in the UZA. We treat observations as missing if the nearest reporting weather station was more than 30 miles from the largest city. Hot days are defined as the days with a high temperature greater than or equal to 30°C.¹¹ Cold days are defined as the days with a low temperature that is less than or equal to 0°C. Rainy days are defined as days with daily precipitation greater than or equal to 1 inch. Finally, snow days are defined as days with on-ground snow accumulations greater than 1 inch. Our measures in each case are the count of days in the month that surpassed these thresholds.

We restrict our sample to UZAs that received Uber by the end of our observation window. Our estimation sample is based on a sliding window of 10 quarters before and after Uber's entry into a location. Cities vary in the types of public transit modes they offer. Whereas virtually every city offers city bus services, a much smaller portion supplies other modes. As a result, fewer UZAs enter our commuter rail estimations than enter our city bus estimations. Given the differing samples, we present descriptive statistics separately for each transit mode. Descriptive statistics for the city bus panel are provided in Table 2, and those for the commuter rail panel are presented in Table 3. We also report sample-specific correlation tables for each mode in Online Appendix D.

Table 2. Descriptive Statistics: City Bus

Statistic	N	Minimum	Maximum	Mean	Standard deviation
Log(Utilization)	14,996	1.115	4.474	2.969	0.483
Log(PassengerTrips)	14,996	7.039	18.489	12.309	1.565
Log(VehicleSupply)	14,996	5.043	14.592	9.340	1.268
RideHailing	14,996	0.000	1.000	0.505	0.500
Log(LaborForce)	14,996	8.781	15.269	11.148	0.995
Unemployment	14,996	1.300	32.100	6.567	2.830
GasPrice	14,996	1.658	4.290	3.031	0.654
NumHotDays	14,456	0.000	31.000	6.023	9.262
NumColdDays	14,455	0.000	31.000	6.814	9.973
NumRainyDays	14,123	0.000	9.000	0.804	1.118
NumSnowDays	14,123	0.000	31.000	2.001	6.093
AvgTripLength	14,728	0.000	25.000	4.733	2.529
ViolentCrime	14,996	0.361	10.882	3.394	1.629
TransitQuality	13,879	0.400	9.600	4.698	1.808

Table 3. Descriptive Statistics: Commuter Rail

Statistic	N	Minimum	Maximum	Mean	Standard deviation
Log(Utilization)	833	1.613	4.552	3.578	0.537
Log(PassengerTrips)	833	9.736	16.967	12.374	1.691
Log(VehicleSupply)	833	6.194	13.144	8.795	1.517
RideHailing	833	0.000	1.000	0.677	0.468
Log(LaborForce)	833	10.837	15.269	12.708	1.008
Unemployment	833	2.450	17.400	7.053	2.695
GasPrice	833	1.658	4.290	3.299	0.636
NumHotDays	813	0.000	31.000	6.387	9.693
NumColdDays	813	0.000	31.000	5.727	9.510
NumRainyDays	833	0.000	8.000	0.701	1.050
NumSnowDays	833	0.000	31.000	2.065	6.361
AvgTripLength	829	14.259	79.666	32.126	16.466
ViolentCrime	833	1.096	8.550	4.181	1.958
TransitQuality	774	3.400	9.600	7.433	1.659

4.2. Regression Specification

Our identification strategy exploits spatial and temporal variation in Uber's entry to estimate a difference-in-differences regression, similar to that of Greenwood and Wattal (2017) and Burtch et al. (2017). We restrict our analysis to cities that are treated within our observation period to ensure that the cities and their transit agencies are reasonably comparable in terms of their probability and timing of treatment and also because this enables an evaluation of the parallel-trends assumption.¹² We focus on Uber in these analyses because Uber was the first entrant in more than 90% of locations. In any case, in locations where Lyft operates alongside Uber, the second provider typically arrives within a handful of months after the first. As such, Uber's entry can be reasonably employed as a proxy for ride-hailing service access in a location generally.

We begin by estimating a separate regression for each public transit mode, with a combination of location and time fixed effects. We employ the natural log of utilization as our outcome measure because the distribution of the variable is heavily right-skewed and because this facilitates elasticity interpretations of the estimated effects.¹³ Our final regression specification is reflected by Equation (1), in which m indexes mode, i indexes UZA, and t indexes time (month). Further, α_i represents our city fixed effects, and γ_t represents our time period fixed effects. Also, $RideHailing_{i,t}$ is the binary indicator of Uber presence.

We initially focus on the main effect of $RideHailing_{i,t}$ and our dynamic controls D , which include $Log(LaborForce_{i,t})$, $GasPrice_{i,t}$, $NumHotDays_{i,t}$, $NumColdDays_{i,t}$, $NumRainyDays_{i,t}$, and $NumSnowDays_{i,t}$. Subsequently, we introduce the moderation effects, between $RideHailing_{i,t}$ and the same dynamic controls, as well as between $RideHailing_{i,t}$ and our static controls S_i , which include $ViolentCrime_i$, $AvgTripLength_i^m$,

and $TransitQuality_i$ (note that the main effects of S are subsumed by our location fixed effects). We first estimate a series of partial specifications, introducing each moderator sequentially and independently, before arriving at the complete specification.

$$\begin{aligned}
 & \text{Log(Utilization}_{i,t}^m) \\
 &= \alpha_i + \gamma_t + \beta_1 \cdot D_{i,t} + \beta_2 \cdot RideHailing_{i,t} \\
 & \quad + \beta_3 \cdot RideHailing_{i,t} \cdot S_i + \beta_4 \\
 & \quad \cdot RideHailing_{i,t} \cdot D_{i,t} + \epsilon_{i,t}.
 \end{aligned} \tag{1}$$

Because all the regressions are log linear, the coefficients can be interpreted as the percent change in utilization on a one-unit change in an independent variable. It is worth noting that some moderators in this regression are not observable in every location at every point in time; thus, introducing some of these moderators reduces the size of the estimation sample.

In addition to the preceding estimations, we also report a set of other tests in the online appendices, with an eye toward exploring the dynamics of these effects (Online Appendix A) and assessing robustness (Online Appendix C considers alternative dependent variables, Online Appendix B reports on a pair of placebo tests, and Online Appendix G considers the incremental effect of Lyft entry). Finally, to help inform policymakers, we also provide our estimates of the gains and losses associated with specific transit agencies and transit modes in Online Appendix I.

5. Results

We begin by estimating the main effects of Uber presence for each transit mode, with the basic difference-in-differences regression reflected by Equation (1). We then introduce the interaction terms, independently and then jointly. As noted earlier, we limit our

Table 4. Difference-in-Differences Regression: Ride-Hailing Impact on City Bus Utilization

DV = Log(Utilization)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log(LaborForce)	0.3746*** (0.0588)	0.3762*** (0.0588)	0.4547*** (0.0594)	0.3728*** (0.0588)	0.3522*** (0.0588)	0.3609*** (0.0592)	0.4069*** (0.0612)	0.3840*** (0.0588)	0.4655*** (0.0629)
Unemployment	−0.0164*** (0.0017)	−0.0164*** (0.0017)	−0.0158*** (0.0017)	−0.0159*** (0.0017)	−0.0172*** (0.0017)	−0.0162*** (0.0017)	−0.0178*** (0.0017)	−0.0165*** (0.0017)	−0.0169*** (0.0018)
GasPrice	−0.0163* (0.0071)	−0.0180* (0.0071)	−0.0195** (0.0071)	−0.0290*** (0.0078)	−0.0184** (0.0071)	−0.0158* (0.0072)	−0.0157* (0.0074)	−0.0135+ (0.0072)	−0.0264** (0.0083)
NumHotDays	−0.0002 (0.0003)	−0.0002 (0.0003)	−0.0002 (0.0003)	−0.0002 (0.0003)	−0.0002 (0.0003)	−0.0002 (0.0003)	−0.0005 (0.0003)	0.0005 (0.0003)	−0.0000 (0.0003)
NumColdDays	0.0009** (0.0003)	0.0009** (0.0003)	0.0010*** (0.0003)	0.0009** (0.0003)	0.0009** (0.0003)	0.0009** (0.0003)	0.0011*** (0.0003)	0.0010** (0.0003)	0.0008* (0.0004)
NumRainyDays	−0.0048*** (0.0013)	−0.0048*** (0.0013)	−0.0048*** (0.0013)	−0.0048*** (0.0013)	−0.0047*** (0.0013)	−0.0049*** (0.0013)	−0.0048*** (0.0013)	−0.0058** (0.0018)	−0.0056** (0.0019)
NumSnowDays	−0.0004 (0.0003)	−0.0004 (0.0003)	−0.0004 (0.0003)	−0.0004 (0.0003)	−0.0004 (0.0003)	−0.0005 (0.0003)	−0.0005 (0.0003)	−0.0001 (0.0004)	−0.0001 (0.0004)
RideHailing		−0.0129* (0.0051)	0.2386*** (0.0318)	−0.0747*** (0.0187)	0.0197** (0.0074)	−0.0338*** (0.0072)	−0.0272** (0.0091)	−0.0054 (0.0064)	0.1282** (0.0401)
RideHailing × Log(LaborForce)			−0.0226*** (0.0028)						−0.0227*** (0.0034)
RideHailing × GasPrice				0.0204*** (0.0060)					0.0289*** (0.0067)
RideHailing × ViolentCrime					−0.0096*** (0.0016)				−0.0031+ (0.0018)
RideHailing × AvgTripLength						0.0051*** (0.0011)			0.0052*** (0.0011)
RideHailing × TransitQuality							0.0032* (0.0016)		0.0039* (0.0018)
RideHailing × NumRainyDays								0.0018 (0.0023)	0.0018 (0.0025)
RideHailing × NumSnowDays								−0.0009 (0.0005)	−0.0011+ (0.0006)
RideHailing × NumColdDays								0.0001 (0.0004)	0.0006 (0.0004)
RideHailing × NumHotDays								−0.0013*** (0.0003)	−0.0010** (0.0003)
N	13,941	13,941	13,941	13,941	13,941	13,673	12,990	13,941	12,722
R ² (Adjusted)	0.2334	0.2337	0.2372	0.2343	0.2357	0.2373	0.2452	0.2347	0.2548
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Employing a wild cluster bootstrapped standard error, as advocated by Cameron et al. (2008), we observe a p -value of 0.007 associated with the effect of ride-hailing in the basic model. Standard errors in parentheses; DV, dependent variable.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

attention to city bus and commuter rail services because these modes bear features that make them, respectively, likely to experience substitution and complementarity—that is, an on-road short-haul service and a rail-based long-haul service.

5.1. City Bus Utilization

We begin by considering the results related to city bus services, which are presented in Table 5. Considering column (2), we observe that the presence of ride-hailing services is associated with a significant decline in city bus utilization, to the tune of 1.29% ($p < 0.05$). Beyond the basic average effect of ride-

hailing, we also observe a variety of significant moderators in columns (3)–(9). Before interpreting these estimates, we remind the reader that the main effect of ride-hailing in these interaction models reflects the impact of *RideHailing* availability conditional on moderators.

We first observe that the relationship between ride-hailing and city bus utilization is positively moderated by the size of the local population (columns (3) and (9) in Table 5) such that a one-unit increase in logged labor force size (a change of nearly 100,000 at the mean) is associated with a 2.3% larger decline in city bus utilization upon Uber's arrival.

Table 5. Difference-in-Differences Regression: Ride-Hailing Impact on Commuter Rail Utilization

DV = Log(Utilization)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log(LaborForce)	1.3974*** (0.3151)	1.4343*** (0.3150)	1.2685*** (0.3421)	1.5399*** (0.3164)	1.2941*** (0.3068)	1.2249*** (0.3569)	1.4679*** (0.3500)	1.4835*** (0.3152)	1.4267*** (0.3756)
Unemployment	−0.0395*** (0.0061)	−0.0405*** (0.0061)	−0.0414*** (0.0062)	−0.0360*** (0.0064)	−0.0309*** (0.0061)	−0.0403*** (0.0061)	−0.0378*** (0.0064)	−0.0401*** (0.0062)	−0.0248*** (0.0068)
GasPrice	0.0489* (0.0224)	0.0506* (0.0224)	0.0496* (0.0224)	−0.0244 (0.0364)	0.0438* (0.0218)	0.0448* (0.0224)	0.0442+ (0.0234)	0.0468* (0.0229)	−0.0591 (0.0397)
NumHotDays	−0.0014+ (0.0007)	−0.0015* (0.0007)	−0.0014* (0.0007)	−0.0015* (0.0007)	−0.0018* (0.0007)	−0.0014+ (0.0007)	−0.0020** (0.0008)	−0.0020* (0.0009)	−0.0018+ (0.0010)
NumColdDays	−0.0011 (0.0008)	−0.0011 (0.0008)	−0.0012 (0.0008)	−0.0012 (0.0008)	−0.0009 (0.0008)	−0.0011 (0.0008)	−0.0007 (0.0008)	−0.0019+ (0.0011)	−0.0023+ (0.0012)
NumRainyDays	−0.0048 (0.0039)	−0.0050 (0.0039)	−0.0050 (0.0039)	−0.0050 (0.0039)	−0.0050 (0.0038)	−0.0051 (0.0039)	−0.0054 (0.0042)	0.0052 (0.0062)	−0.0003 (0.0069)
NumSnowDays	0.0006 (0.0009)	0.0007 (0.0009)	0.0006 (0.0009)	0.0006 (0.0009)	0.0003 (0.0009)	0.0006 (0.0009)	0.0005 (0.0009)	0.0036** (0.0013)	0.0018 (0.0014)
RideHailing		0.0296* (0.0148)	−0.1642 (0.1571)	−0.2874* (0.1228)	−0.0812*** (0.0217)	0.0517* (0.0234)	0.1453** (0.0448)	0.0402* (0.0181)	−0.3710 (0.2769)
RideHailing × Log(LaborForce)			0.0154 (0.0124)						−0.0052 (0.0156)
RideHailing × GasPrice				0.0876** (0.0337)					0.1113** (0.0381)
RideHailing × ViolentCrime					0.0263*** (0.0039)				0.0253*** (0.0052)
RideHailing × AvgTripLength						−0.0007 (0.0006)			−0.0004 (0.0008)
RideHailing × TransitQuality							−0.0162** (0.0058)		−0.0033 (0.0079)
RideHailing × NumRainyDays								−0.0164* (0.0074)	−0.0089 (0.0082)
RideHailing × NumSnowDays								−0.0050** (0.0016)	−0.0026 (0.0018)
RideHailing × NumColdDays								0.0012 (0.0012)	0.0021 (0.0013)
RideHailing × NumHotDays								0.0006 (0.0009)	−0.0007 (0.0010)
N	813	813	813	813	813	809	754	813	750
R ² (Adjusted)	0.2277	0.2308	0.2313	0.2365	0.2737	0.2329	0.2433	0.2429	0.2864
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Employing a wild cluster bootstrapped standard error, as advocated by Cameron et al. (2008), we observe a p -value of 0.03 associated with the effect of ride-hailing in the basic model. Standard errors in parentheses; DV, dependent variable.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

We further observe that the decline in city bus utilization is significantly stronger when gas prices are reduced (columns (4) and (9) in Table 5) such that a dollar decrease in gas prices is associated with a 2%–3% larger drop in city bus utilization upon Uber's arrival. Considering *ViolentCrime* (columns (5) and (9) in Table 5), we observe that one additional violent crime per thousand is associated with a 1%–3% larger drop in city bus utilization upon Uber's arrival.

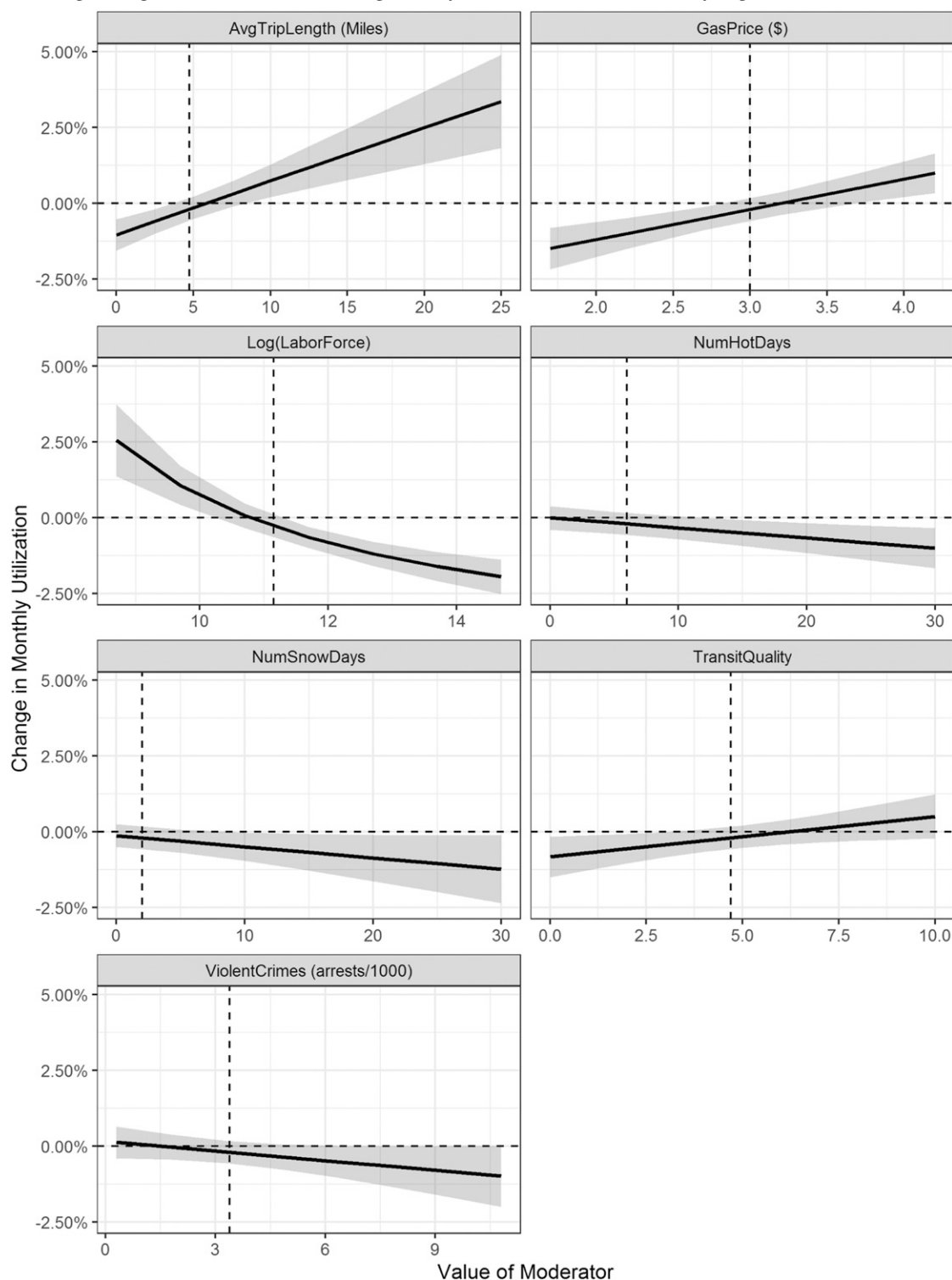
We also observe that ride-hailing has a significantly more negative effect on city bus utilization when the typical rider trip distance is shorter (columns (6) and (9) in Table 5). Specifically, a 1-mile

increase in average trip distance is associated with an ~0.5% larger drop in city bus utilization upon Uber's arrival. The effects also depend on the perceived quality of transit services such that a one-point decrease in transit quality ratings (columns (7) and (9) in Table 5) is associated with a 0.3%–0.4% larger drop in city bus utilization upon Uber's arrival. Finally, we observe that inclement weather leads to a significantly more negative effect on city bus utilization (columns (8) and (9) in Table 5), particularly in the case of heat. We find that one additional hot day in a given month is associated with a 0.1%–0.13% larger drop in city bus utilization upon Uber's arrival.

We further observe that a greater volume of snowy days is associated with a stronger drop in city bus utilization when Uber arrives (this becomes clear in column (9), our complete moderation model). Our estimates suggest that having one additional day of high snow cover is associated with a 0.1%

stronger decline in city bus utilization upon Uber's arrival. These various moderating effects are depicted visually in Figure 1, where we plot average marginal effects of *RideHailing* on city bus utilization over the distribution of each statistically significant moderator.

Figure 1. Average Marginal Effect of Ride-Hailing on City Bus Utilization, Statistically Significant Moderators



5.2. Commuter Rail Utilization

We next consider the results related to commuter rail, which are presented in Table 6. First, we see that ride-hailing is associated with a significant increase in the utilization of commuter rail services. That is, upon Uber's arrival, commuter rail utilization increases by approximately 3%. Second, considering sources of heterogeneity (columns (3)–(9) in Table 6), we once again see evidence of some moderating effects. Specifically, we observe that the relationship between Uber's presence and commuter rail utilization is moderated by *GasPrice* (columns (4) and (9) in Table 6) such that a dollar increase in the gas price is associated with an ~8%–11% increase in commuter rail utilization following Uber's arrival. Further, we observe that the effect of Uber on the utilization of commuter rail services is also moderated by local rates of violent crime (columns (5) and (9) in Table 6). Specifically, our estimates indicate that one additional crime per thousand people is associated with an ~2.5% larger increase in commuter rail utilization following Uber's arrival.

In contrast to the city bus results, we do not see consistent evidence that the relationship between ride-hailing and the utilization of commuter rail is moderated by the size of the local population, riders' average trip distances, transit quality, or inclement weather. Of course, the absence of statistically significant effects does not constitute evidence to the contrary. We may lack statistical power to precisely quantify these relationships. Moreover, we may face some difficulty detecting moderating effects by construction because of a lack of variation on the measures—namely average trip distance and population size. This is because commuter rail services are typically only available in larger cities and because even the

shortest passenger trips are generally still long on commuter services by definition. This lack of variation in the sample may mask such relationships. Once again, we plot these moderating effects graphically in Figure 2, where we visualize the average marginal effect of *RideHailing* on commuter rail utilization over the distribution of each statistically significant moderator.

Besides these results, we also estimated similar effects for other common transit modes—namely commuter bus, light rail, and subway services. We found no significant average effects of ride-hailing on the monthly utilization of these modes. However, we did find that crime and weather significantly shaped the impact of ride-hailing on both light rail and subway use. The results reflecting this can be found in Online Appendix E.

6. Discussion and Implications

We summarize our findings related to heterogeneity around the effect of ride-hailing on city bus and commuter rail utilization in Table 6. Based on our findings, transit agency operators contemplating partnerships with ride-hailing services should consider the losses or gains they have experienced (or might experience going forward) deriving from the arrival of ride-hailing services. In the event of a loss, those losses should be compared against the cost of any subsidies the agency would consider providing to riders to facilitate their use of ride-hailing while also considering potential savings from discontinuing transit routes or reducing service frequency. Transit agencies and ride-hailing services are likely to benefit from collaboration when it comes to modes that maintain distinct ROWs, that service long-distance travelers, that operate at generally higher quality, that service

Figure 2. Average Marginal Effect of Ride-Hailing on Commuter Rail Utilization, Statistically Significant Moderators

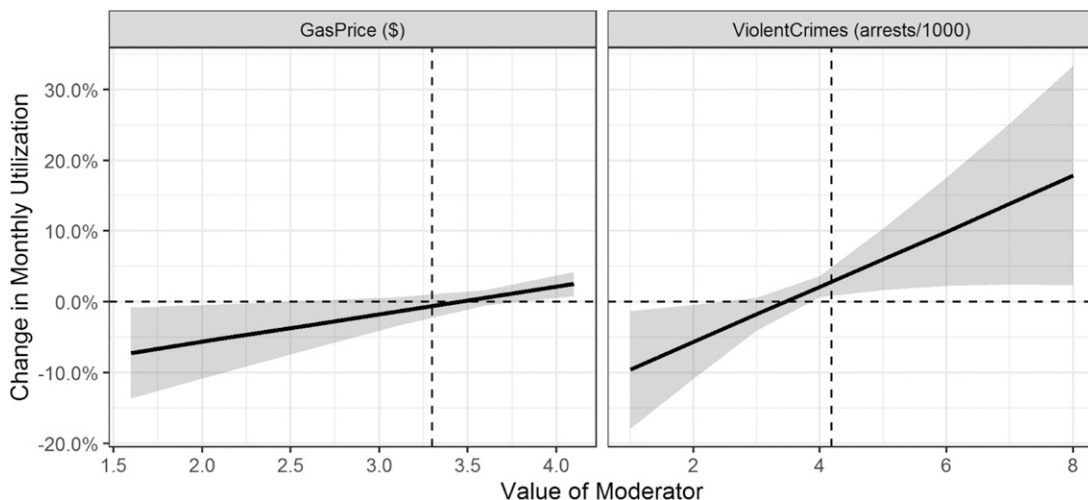


Table 6. Summary of Results

Average effect	City bus	Commuter rail
<i>RideHailing</i>	↓	↑
Moderating effects		
<i>Log(LaborForce)</i>	↓	—
<i>GasPrice</i>	↑	↑
<i>ViolentCrime</i>	↓	↑
<i>AvgTripLength</i>	↑	—
<i>TransitQuality</i>	↓	—
<i>NumRainyDays</i>	—	—
<i>NumSnowDays</i>	↓	—
<i>NumColdDays</i>	—	—
<i>NumHotDays</i>	↓	—

more populous regions, or that operate under frequent inclement weather. A typical collaboration would take the form of an incentive, which may be implemented as direct subsidization of consumers' use of ride-hailing, joint passes, or guaranteed ride programs. Notably, such programs are already being contemplated by several transit agencies.

The majority of partnerships between public transit agencies and ride-hailing services that have been initiated thus far have largely been tied to city bus services. However, our findings suggest that agencies may stand to benefit even more from engaging in partnerships that expand access to commuter rail services, where we have observed evidence of complementarity. Moreover, it is important for transit operators and municipal governments to account for the local context. Our findings indicate that the potential for complementarity or substitution varies to some degree even within a given mode of transit such that complementarity may manifest with city bus services under the right conditions—for example, in smaller cities, when gas prices are elevated, when transit quality is perceived to be high, when commuters tend to use bus services to travel lengthy distances, when crime is relatively low, and when temperatures are not excessive. Similarly, we observe that the complementarity between ride-hailing and commuter rail services tends to be weaker when gas prices and crime are relatively low. The effect estimates we observe may appear relatively small in size, but they translate to substantial annual losses per transit agency.¹⁴ Of course, when considering the privatization of any public good, transit providers should take care and note that ride-hailing services do not necessarily share the same incentives when it comes to transit provision. Whereas public agencies are charged with delivering equitable services to all residents and socioeconomic groups of a city, transit providers are for-profit companies (note that both Lyft and Uber have recently gone public). Accordingly, any partnerships likely should also involve pricing agreements, regulation, and municipal reporting requirements.

By contrast, our findings have important implications for ride-hailing providers not only in seeking out collaborations with public transit operators but also in planning any further expansions into new geographies. Our results speak to specific travel needs that are frequently not satisfied by existing public transit providers, which constitute gaps and inefficiencies that ride-hailing services may accommodate and address.

7. Limitations and Conclusion

To analyze Uber's impact across the United States, we needed to rely on monthly observations of transit use, aggregated to the agency-location level. This means that we are not able to arrive at a nuanced understanding of within-city daily dynamics in transit use, nor are we able to draw conclusive claims about the exact mechanisms driving reductions and expansions in transit mode use in any given instance. We see two opportunities to resolve this chief limitation of our work while maintaining broad, generalizable insights. One promising opportunity would be to examine trip-level data from smart cards and transit passes, which track unique riders across modes (Pelletier et al. 2011). Many jurisdictions in Europe and Asia employ such systems, and they are beginning to make their way to North America as well—for example, Toronto's PRESTO card. Second, future work might look to tackle similar questions with a similar design but employing complementary data sources—for example, data on ride-hailing supply (Jiang et al. 2018) and demand (Park et al. 2016) and how such measures respond to variation in the availability of public transit options. This amounts to a very similar research design, except that the focus would be on how changes to public transit shock ride-hailing demand. Analyses of this sort could complement the analysis we present in this study, lending greater credibility and/or more nuanced understanding. Hoffman et al. (2016) provide an early example of the approach we have in mind. They study the effect of subway station closures on the demand for ride-hailing in nearby locations. More generally, we see opportunities for future work to expand the scope of this analysis to incorporate additional travel modes, such as taxi or bike share. Although we lack the necessary data, some emerging work has already begun to explore these travel alternatives (Jin et al. 2018). Our work thus constitutes an early effort in this space, and we hope that subsequent work will expand this line of inquiry to further collective understanding of the implications that derive from novel forms of technology-enabled transit options.

Endnotes

¹ See, e.g., Lyft help: Commuter benefits, <https://help.lyft.com/hc/en-us/articles/115013080828>; Government subsidies for Uber and

Lyft fares, <https://www.businessinsider.com/some-cities-are-subsidizing-uber-rides-to-cut-public-transportation-costs-2016-12>.

² Results for other modes can be found in Online Appendix E.

³ In Online Appendix H, we report another pertinent analysis, considering the fraction of transit routes operated by a city bus service or a light rail service that maintains an exclusive ROW (i.e., distinct bus lanes or fixed rail) versus not (i.e., regular vehicle lanes or rail laid atop road). We limit this analysis to the online appendices because these measures are optionally reported by agencies and available only for a small subset.

⁴ We focus on these two transit modes for the sake of exposition. Analyses and discussion of ride-hailing's effects on other common public transit modes—namely light rail, subway, and commuter bus—are provided in Online Appendix E.

⁵ See Uber's fuel finder, <https://www.uber.com/blog/fuel-finder>.

⁶ See, e.g., What should rideshare drivers do about high gas prices? at <https://maximumridesharingprofits.com/uber-drivers-high-gas-prices>.

⁷ Note that we report analyses of other transit modes in the online appendices.

⁸ See, e.g., King County Metro's measure of productivity, at <http://metro.kingcounty.gov/am/future/pdfs/20140409-metro-productivity-measures.pdf>.

⁹ See the U.S. Energy Information Administration website for more details: https://www.eia.gov/petroleum/gasdiesel/gas_proc-methods.php.

¹⁰ See all city ratings here: <https://alltransit.cnt.org/rankings>. The detailed method for transit performance score is published by AllTransit here: <http://alltransit.cnt.org/methods/AllTransit-Methods.pdf>.

¹¹ Most transit agencies use heating, ventilation, and air-conditioning systems that trigger air conditioning at 24°C, so our current cutoff is a conservative estimate of what is considered *too hot to walk/wait at transit stops*.

¹² Including untreated cities into a lead-lag regression is not feasible because the lead and lag dummies are undefined in the absence of a treatment date.

¹³ Note that utilization is never less than one because unlinked passenger trips in our sample are strictly greater in number than the volume of vehicle revenue hours.

¹⁴ We provide agency-mode-specific estimates of losses and gains that have been experienced as a result of ride-hailing services in Online Appendix I.

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